Capstone Team Presentations

Group 1: Portfolio Summary and Advisor Book

- The team reported building a phase-one model with staging and integration models, final outputs, reconciliation passes, data-quality checks, build tests, and the required Airflow DAGs.
- Their source data included trading data, CRM, fund references, security, and portfolio management.
- Their consumer-facing product was described as supporting advisory leadership and portfolio oversight.
- The team reported seven source-aligned tables and approximately nine ETL models. The presentation also referred to seven sources; these counts may reflect different layers of the design.
- The two selected outputs were **Portfolio Summary** and **Advisor Book**, intended to help users assess portfolio health, including assets under management.
- The team reported running two jobs and said the Airflow jobs, Databricks catalogs, and resulting data could be inspected.

“We basically tackled ... Portfolio summary and advisor book.”

Data-quality observation

- The team found trades dated before a portfolio’s inception date.
- They chose to retain those records and notify business users, since they did not know whether the dates indicated bad data or a valid business situation.
- A data-quality check could be added if business users identify the records as an issue; otherwise, a business explanation may justify keeping them.

“We decided, hey, let's let our business users know that this is happening, and we will just keep it in there.”

Group 2: Instrument Activity and Client Segments

- This team focused on two aggregate business outputs:
  - **Instrument Activity**: a view of trading activity grouped by instrument classification, intended to help leadership understand which instruments or

asset classes are traded most.
    - **Client-segment performance**: an aggregate intended to compare activity across client segments.
- The team described a pipeline that takes raw seed data through staging and domain-specific processing to consumer-aligned outputs.
- Transformations considered included retaining valid trades, handling price conversion, and aggregating the data.
- Potential issues they identified included:
    - Whether the latest price is fair and current.
    - Invalid trades, fee calculations, and reconciliation.
    - Missing currency conversion, which could make traded-value comparisons inaccurate.
    - Trades referencing unknown instruments, which could be excluded by a join.
    - Duplicate counting if the same trade appears more than once.
    - Inconsistent or incorrect segment labels.
- For positions, the team described calculating net buys and sales for each portfolio and instrument. The transcript's wording is unclear about the exact formula.

Group 3: Portfolio Summary and Advisor Book

- This team also selected **Portfolio Summary** and **Advisor Book**, describing the goal as a trustworthy view of portfolios and advisor books.
- They worked backward from the outputs to identify source data. Sources mentioned included advisor and CRM data, trades, and fund summary or report data.
- The team described using seven seed datasets to produce consumer reports and presented an ERD to support team discussion and PRD development.
- They said they clarified definitions that were unclear in the requirements and identified data-quality validation needs.
- Their presentation also called out data ownership, approvals, and validation.
- Databricks was proposed as the platform for the dbt work.

Additional Team Documentation and Technical Reflections

- Another presenter described creating a summary document using a technique learned earlier in the course.
- The documentation covered the PRD, data contract, lineage diagrams, and calculations, with the work focused on the first two outputs: Portfolio Summary and Advisor Book.
- Basis-point calculations were identified as a point requiring investigation: the team checked whether to divide by 1,000 or 10,000.
- The presenter said the generated documentation included guidance on what readers should not assume.
- Imports were described as the hardest part of phase two, and the presenter said they had stopped there.
- The presenter also described using AI to produce a one-page summary of the day’s work.

Shared Context

- Teams described using AI-assisted development and documentation tools, including Catalyst.
- Several teams produced similar outputs, especially Portfolio Summary and Advisor Book, while another team focused on instrument activity and client-segment aggregates.
- The presentations included both reported implementation work and potential data risks; details such as source counts and some calculations were not fully clear in the transcript.